

Digital THERMOMETER

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Thermometers have been useful for temperature measurement since a long time. A digital thermometer is an electronic device used for accurate temperature measurement. Unlike traditional analogue thermometers, which use mercury or other liquids, digital thermometers utilise electronic sensors to detect and display temperature readings.

Digital thermometers are commonly employed in healthcare settings for monitoring body temperature. They are also valuable in the food industry to ensure proper cooking and storage temperatures. Additionally, they are used in environmental monitoring to measure ambient temperatures in various settings.

Digital thermometers offer convenience, accuracy, and ease of use, making them a popular choice for a wide range of temperature measurement applications. The author's prototype of a digital thermometer on a breadboard is shown in Fig. 1.

Circuit and working

The circuit diagram of the thermometer is shown in Fig. 2. The thermometer is built around an Arduino board (BOARD1), a temperature sensor LM35 (IC1), a 16x2 LCD (LCD1), and a few other components. Simply connect a 9-volt battery or adaptor to the board to operate it.

The LM35 is an analogue temperature sensor that outputs a voltage proportional to the temperature. The Arduino processes this information to display the tempera-

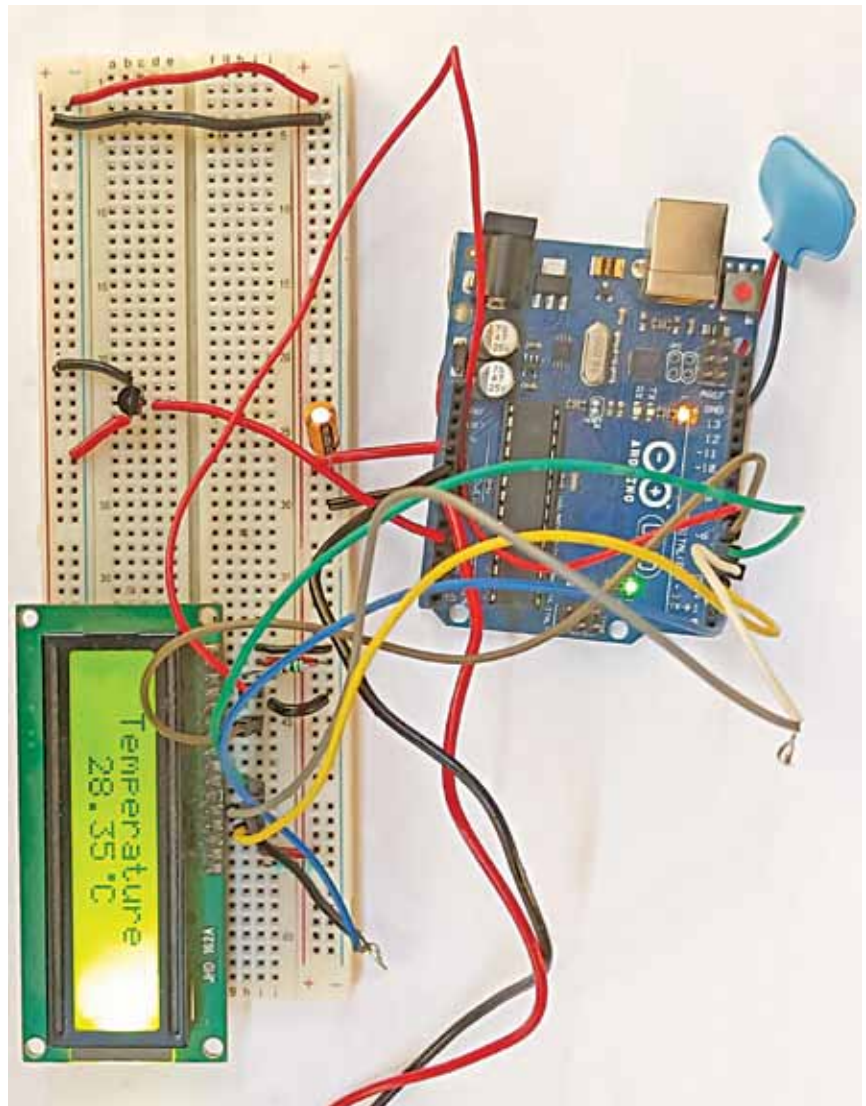


Fig. 1: Author's prototype

ture on the LCD.

The LM35 temperature sensor is used to sense the environment's temperature, producing a signal that indicates 1 degree temperature change for every 10mV change at its output pin. For example, if the output voltage of the LM35 sensor is

250mV, it means the temperature is around 25 degrees Celsius.

Arduino reads this output voltage of the temperature sensor using Analogue pin A1 and performs calculations to convert this analogue value to a digital value representing the current temperature. After these